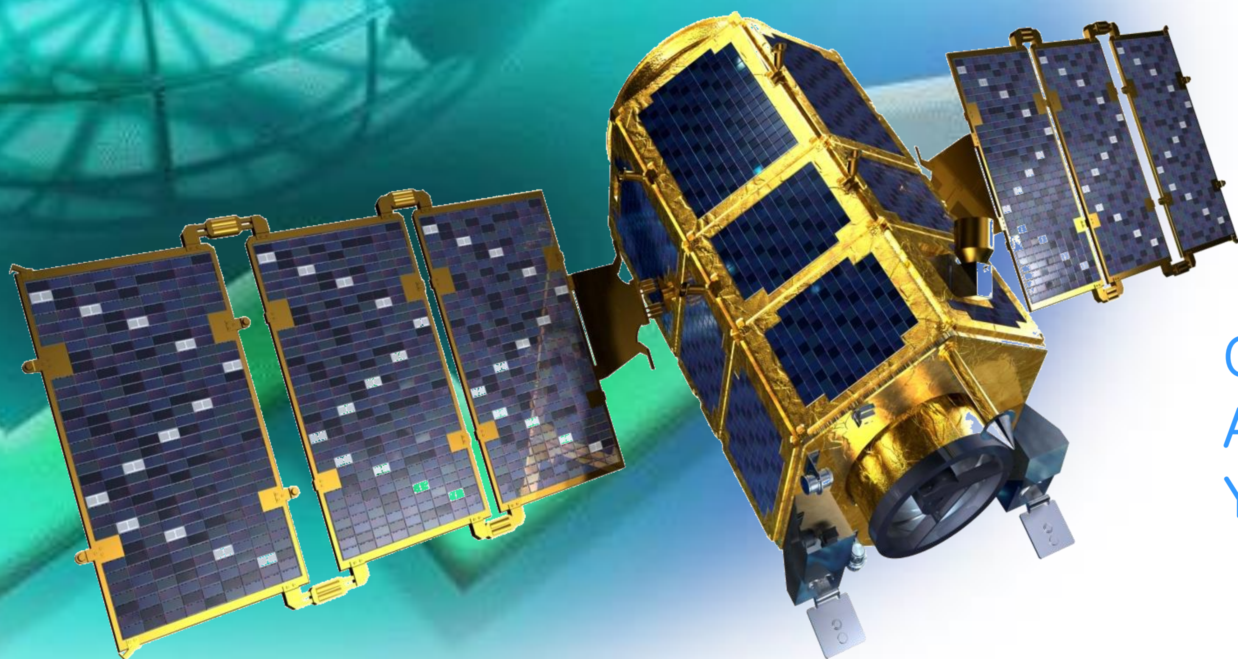
Fall 2011

Spacecraft Systems



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Chapter 10
SPACECRAFT DESIGN & SIZING

10.1 Requirements, Constraints, and the Design Process

10.2 Spacecraft Configuration

10.3 Design Budgets

10.4 Designing the Spacecraft Bus

10.5 Integrating the Spacecraft Design

10.6 Examples

◆ Overview of S/C Design and Sizing (Table 10-1)

TABLE 10-1. Overview of Spacecraft Design and Sizing. The process is highly iterative, normally requiring several cycles through the table even for preliminary designs.

Step	References
1. Prepare list of design requirements and constraints	Sec. 10.1
2. Select preliminary spacecraft design approach and overall configuration based on the above list	Sec. 10.2
3. Establish budgets for spacecraft propellant, power, and weight	Sec. 10.3
4. Develop preliminary subsystem designs	Sec. 10.4
5. Develop baseline spacecraft configuration	Secs. 10.4,10.5
6. Iterate, negotiate, and update requirements, constraints, design budgets	Steps 1 to 5

10.4 Designing the S/C Bus

10.4.1 Propulsion Subsystem

- Propulsion Equipment for a S/C includes:
  - Tankage to hold the propellant
    - Placed near the S/C center of mass
  - Lines and Pressure-Regulating Equipment
  - Engines (or Thrusters)
- Propellant
  - (Cold) Pressurized Gas: Nitrogen
  - Monopropellant: Hydrazine (N_2H_4)
  - Bipropellant
- Important Design Parameters
  - Number/Orientation/Location of the Thruster
    - = Thrust Level
    - = Amount of Impulse Requirement

- Engines for Orbit Control
  - aligned to thrust through the S/C center of mass
- Engines for Attitude Control
  - mounted as far away from the center of mass as possible to increase the lever arm and thus increase the torque per unit thrust
  - Three-axis control requires 6 attitude control thrusters

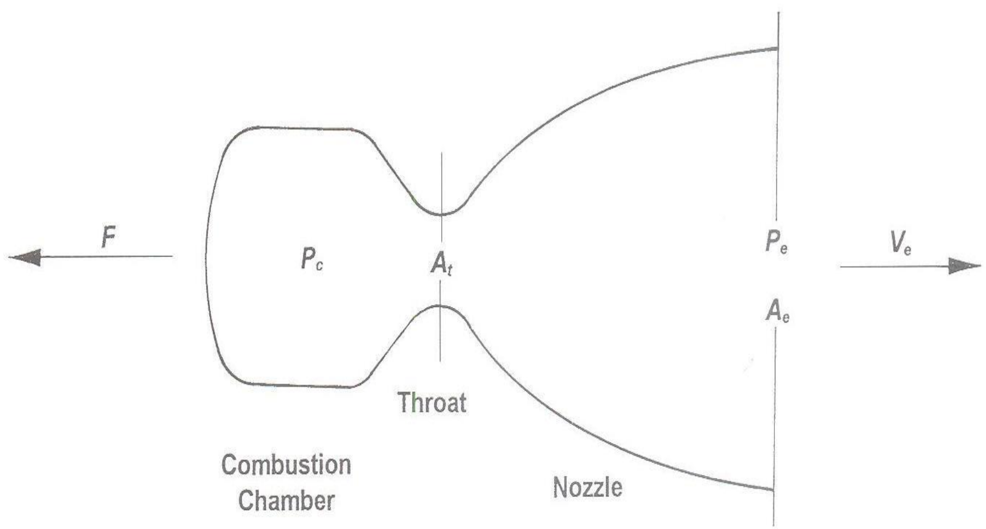


Fig. 4.1 Rocket nozzle.

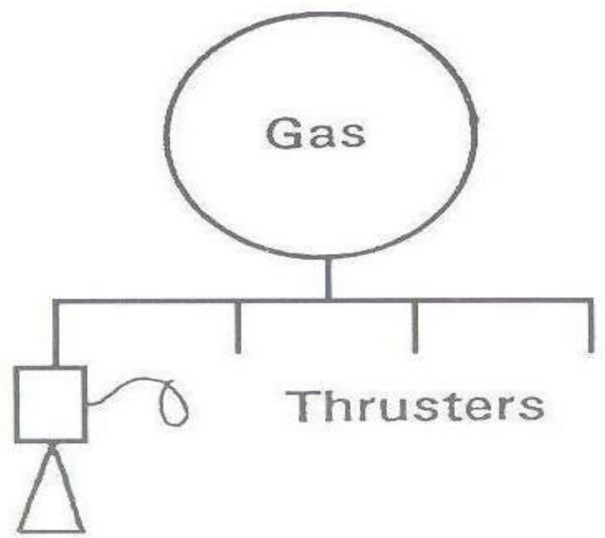


Fig. 4.4 Cold-gas system.

10.4 Designing the S/C Bus

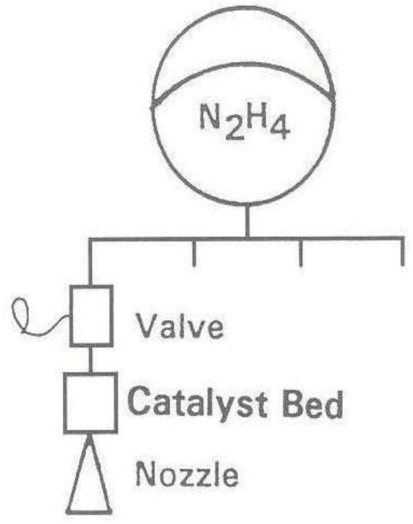


Fig. 4.5 Monopropellant system.

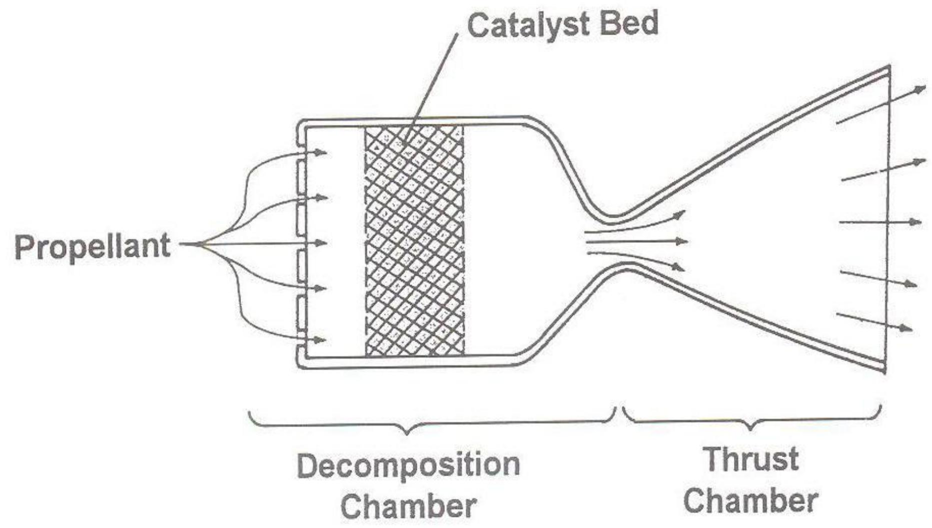


Fig. 4.17 Monopropellant thruster concept.

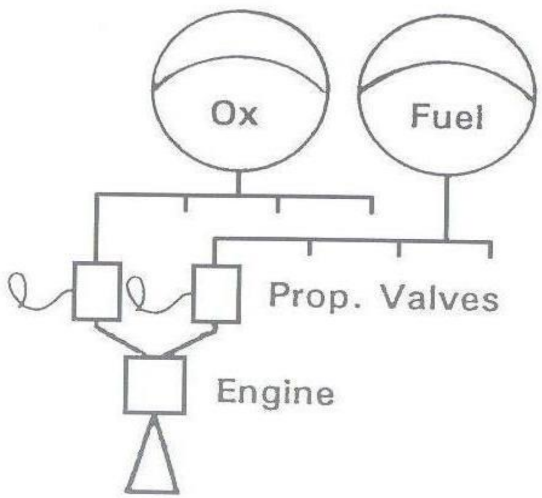


Fig. 4.6 Bipropellant system.

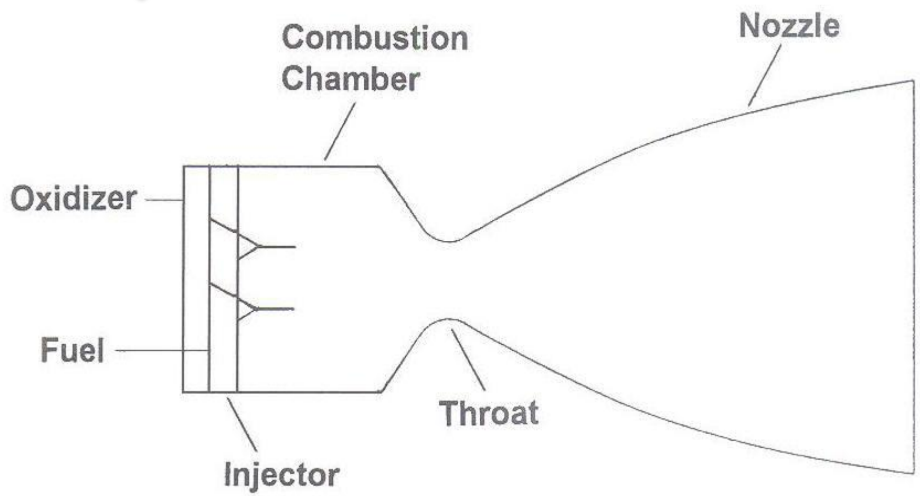


Fig. 4.38 Bipropellant rocket engine.

10.4 Designing the S/C Bus

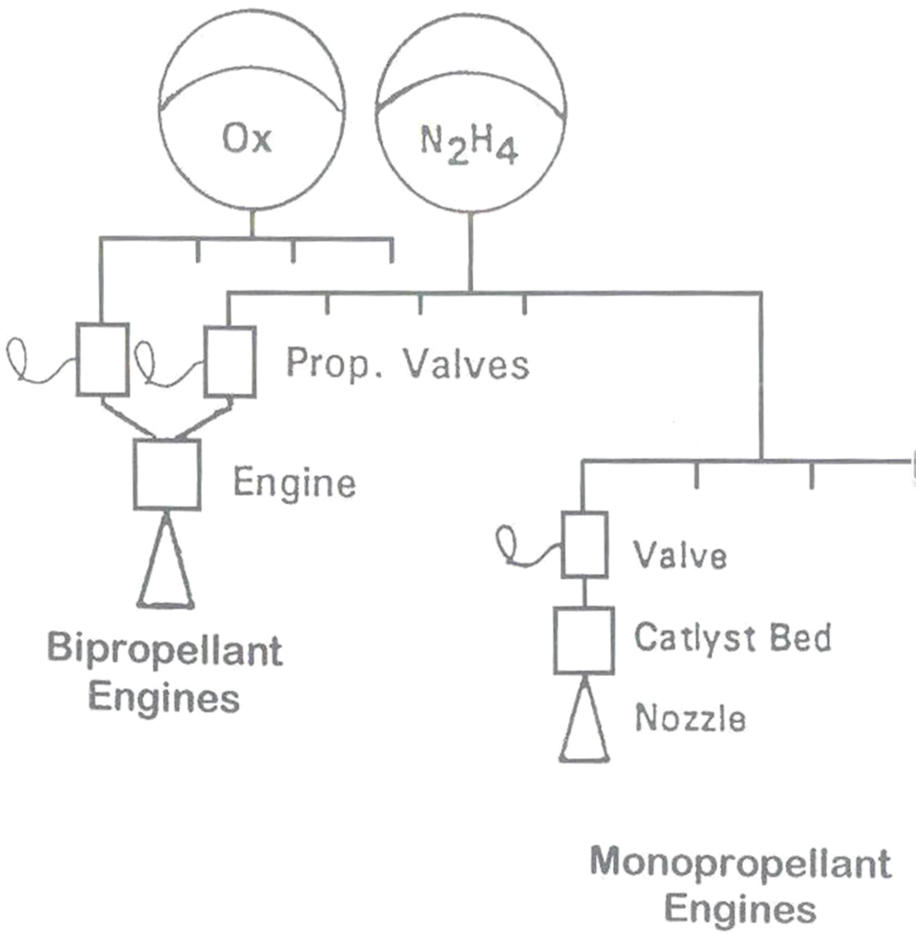


Fig. 4.7 Dual-mode system.

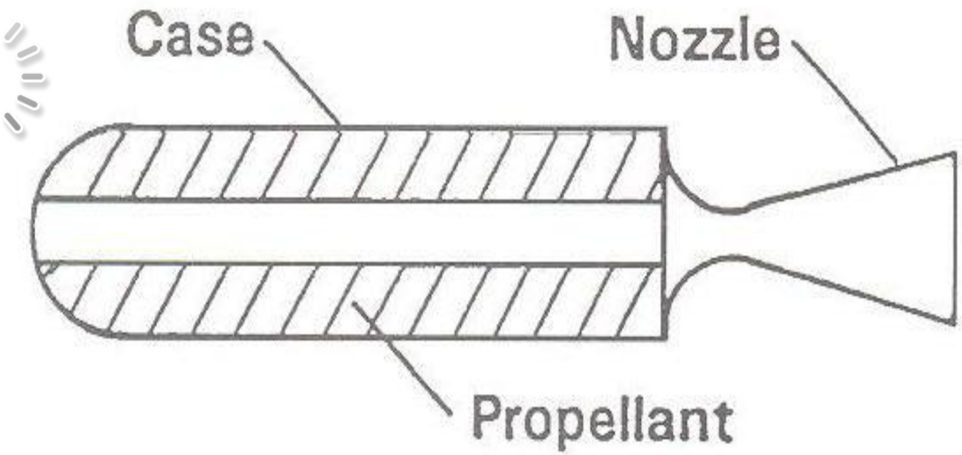


Fig. 4.8 Solid motor.

10.4 Designing the S/C Bus

TABLE 10-12. Weight and Power Budget for Propulsion Subsystem. See Sec. 17.4 for specific design information.

Component	Weight (kg)	Power (W)	Comments
Propellant	Table 10-7	—	Added to overall budget in Table 10-7; **not** part of propulsion subsystem
Tank	10% of propellant weight		Tanks for compressed gas may be up to 50% of gas weight
Thrusters	0.35–0.4 kg for 0.44 to 4.4 N hydrazine units	5 W per thruster when firing	
Lines, Valves, & Fittings	Dependent on detailed spacecraft design	—	Example spacecraft of Sec. 10.6 used 6.8 kg (HEAO) & 7.5 kg (FLTSATCOM)

10.4.2 Attitude Determination & Control Subsystem

- Attitude Control Requirements are based on those for:
  - Payload Pointing
  - S/C Bus Pointing

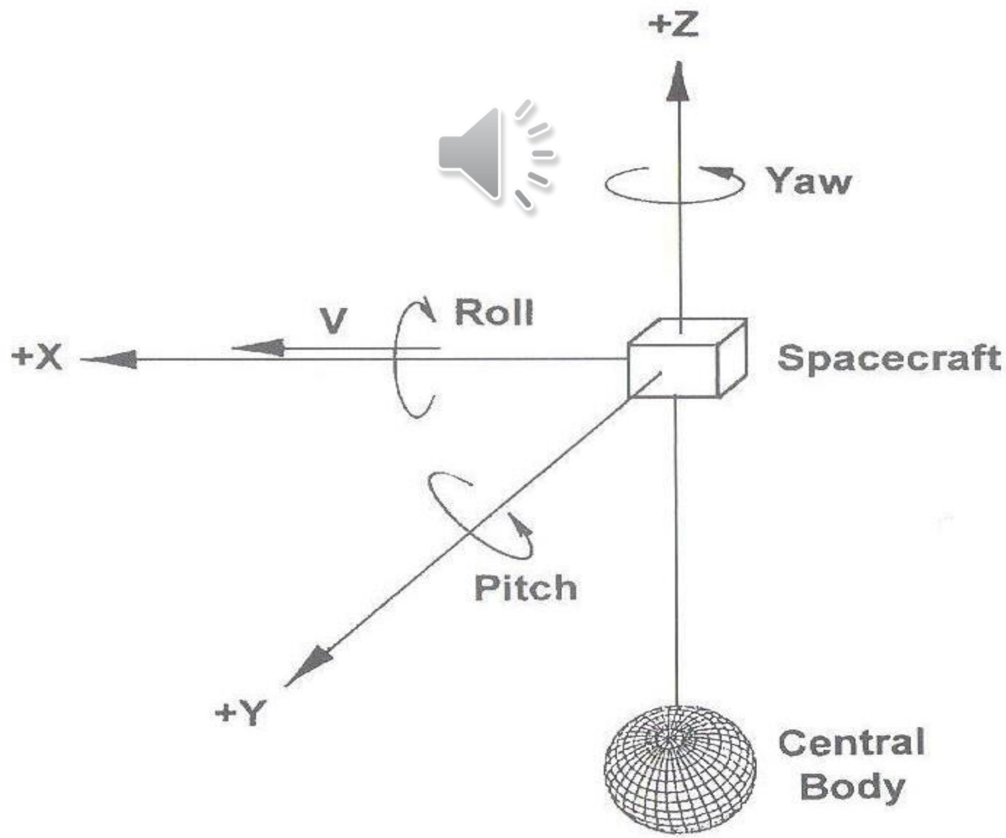


Fig. 5.1 Attitude orientation of a spacecraft.

10.4 Designing the S/C Bus

TABLE 10-13. Typical Sources of Requirements for Attitude Control.

Requirement	Information Needed
***Payload Requirements*:**	
Article to be Pointed	Entire payload or some payload subset such as antennas or a thermal radiator
Pointing Direction	Defined relative to what reference
Pointing Range	All of the possible pointing directions
Pointing Accuracy	Absolute angular control requirement
Pointing Knowledge	Knowledge of pointing direction either in real time or after the fact
Pointing Stability	Maximum rate of change of angular orientation
Slew Rate	Reorientation from one pointing direction to another in a specified time
Exclusion Zones	For example, "not within 10 deg of the Sun"
Other Requirements:	
Sun Pointing	May need for power generation or thermal control
Pointing During Thrusting	May need for guidance corrections
Communications Antenna Pointing	Toward a ground station or relay satellite

10.4 Designing the S/C Bus

TABLE 10-14. Design Approaches for Selected Pointing Requirements.

Requirement	System
Nadir-Pointed Payload	Body-fixed payload using 2 axes of body attitude control to meet the Earth-pointing requirement. The third axis is used to point a horizontal axis in the direction of flight. Can also use spin-stabilized spacecraft with spin axis normal to the orbit plane and payload mounted on despun platform.
Payload Pointed in a Fixed-Inertial Direction	Body-fixed payload using 2 axes of body attitude for payload-pointing direction in inertial space. Third axis is used to keep one side toward the Sun.
Sun-Oriented Solar Array	Planar array requires 2 axes of control. May be achieved by 1 axis of body attitude and 1 rotation axis. Cylindrical array with array axis perpendicular to Sun line.
Communications Antenna	2-axis mechanism.

10.4 Designing the S/C Bus

- Spin Stabilization
  - Use Gyroscopic Effects
  - Cylindrical Configuration in general
  - Extensively used for Kick-Stage Firing and Small S/C

- Spin Stabilization
  - Passive Spin
  - Spin with Precession Control
  - Dual Spin

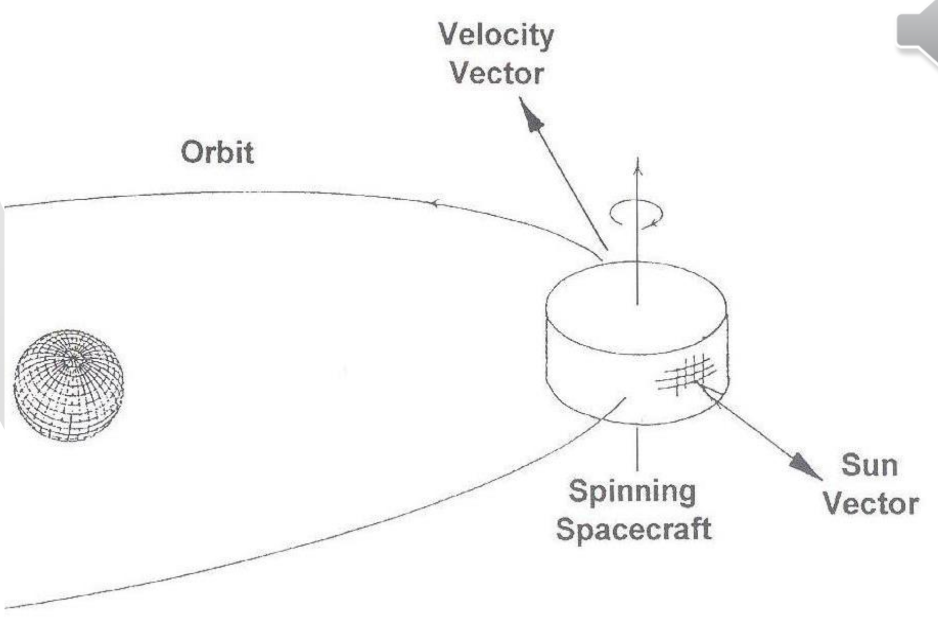


Fig. 5.2 Spinning spacecraft.

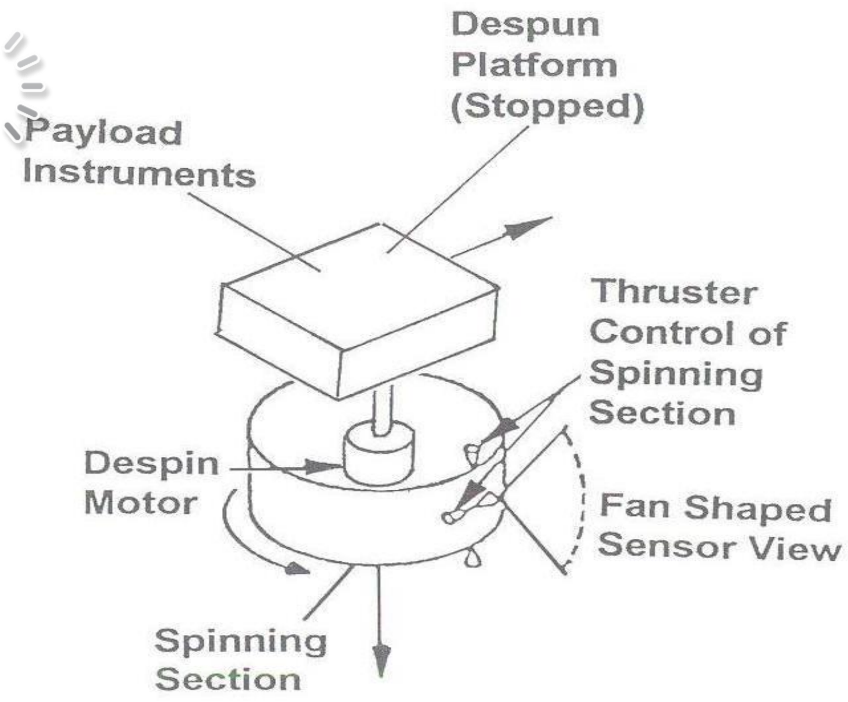


Fig. 5.6 Dual spin.

10.4 Designing the S/C Bus

- Three Axis Control
  - Use Conservation of Angular Momentum
  - Passive vs. Active Control
    - Low Accuracy vs. High Accuracy
    - Clean & Simple vs. Heavier & Complex
    - Usually for Small S/C vs. for Small/Large S/C

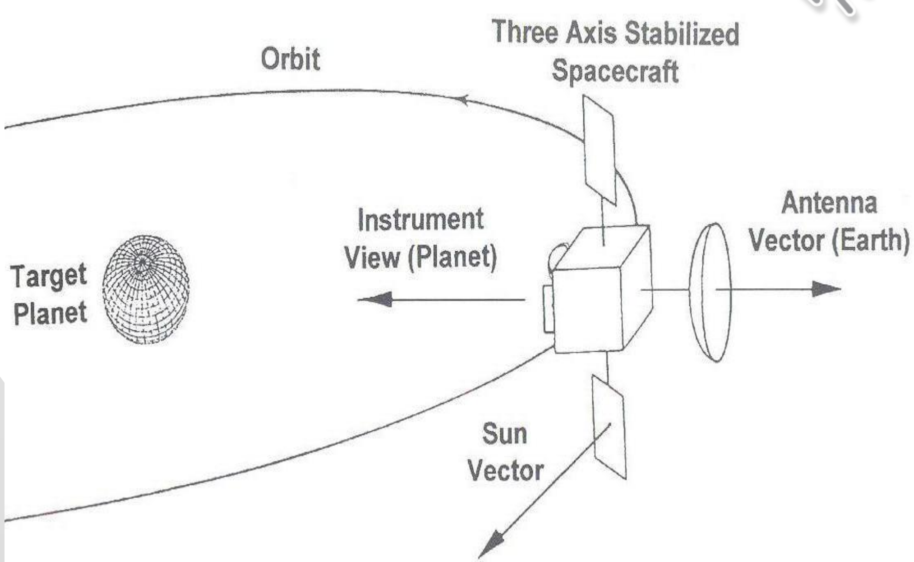


Fig. 5.3 Three-axis-stabilized spacecraft.

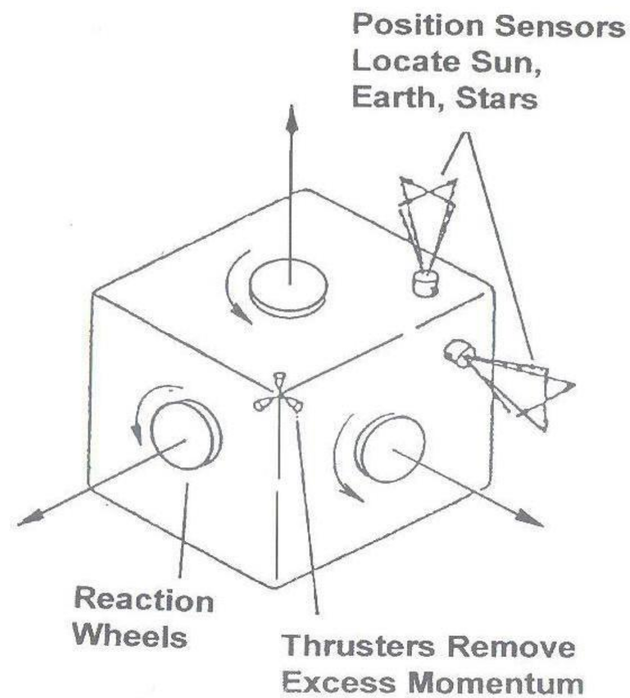


Fig. 5.7 Three-axis stabilized.

10.4 Designing the S/C Bus

- Passive Control (Methods)
  - Gravity-Gradient
  - Magnetic Torque

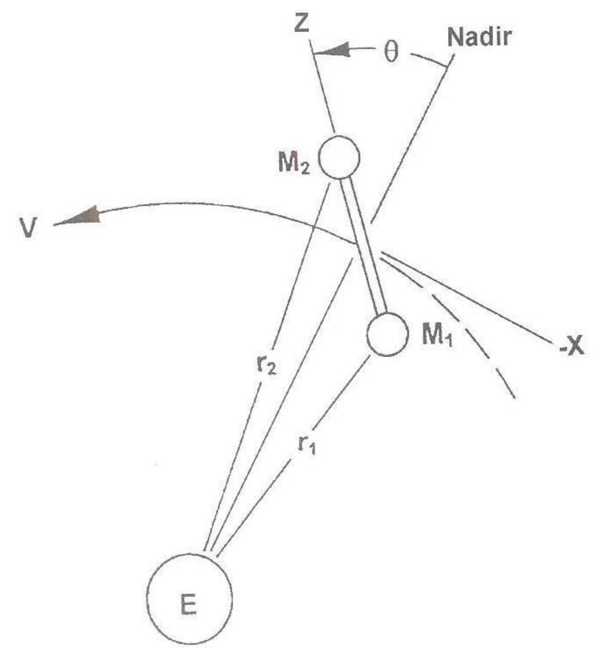
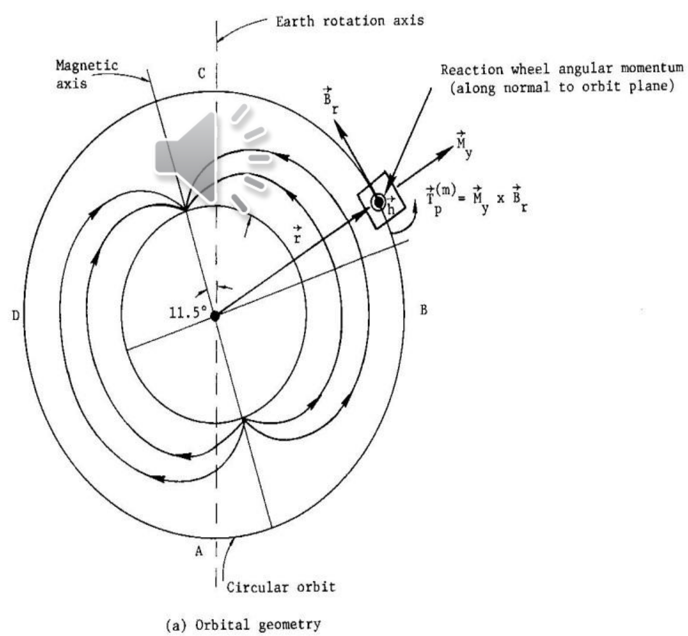
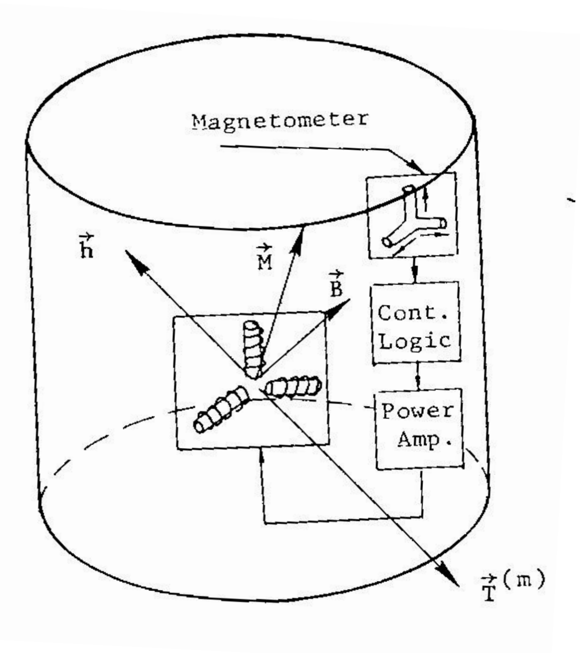


Fig. 5.11 Gravity-gradient torque.



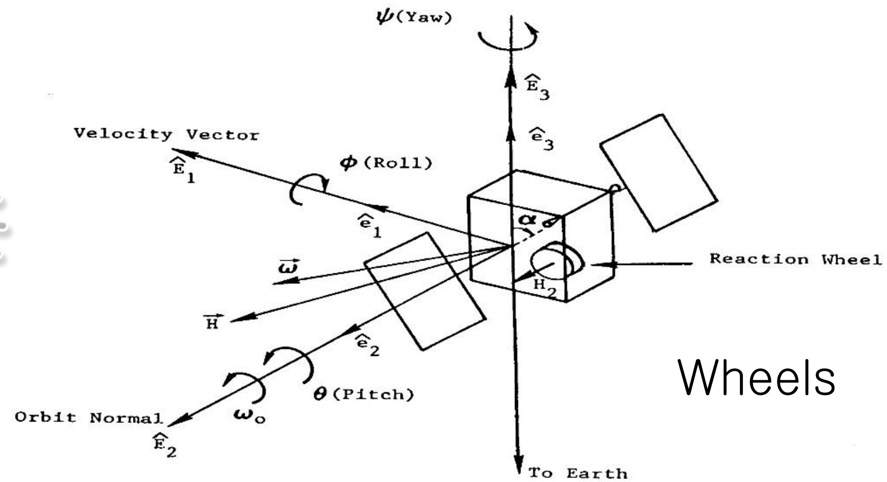


Magnetic Torquer

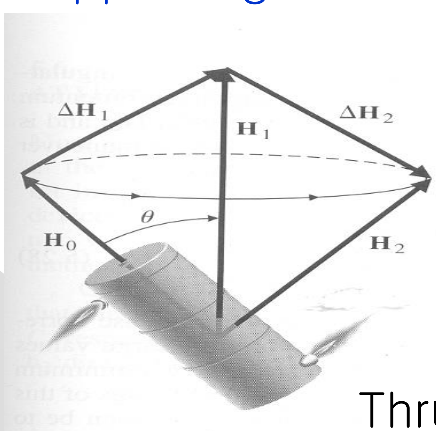
10.4 Designing the S/C Bus

- Active Control (Actuators)
  - Thrusters
    - Large Torque/Large Fuel
  - Wheels
    - Good for Cylindrical Disturbances
    - Less Propellant
    - Momentum/Reaction Wheel
    - Torque < 1Nm
  - Control Moment Gyros
    - Torque < 10^3 Nm
    - More Efficient Solar Arrays Pointing
    - Several Payload or Appendages

- Active Control (Sensors)
  - Earth/Sun/Star
  - Gyroscopes
  - Magnetometers



Wheels



Thruster

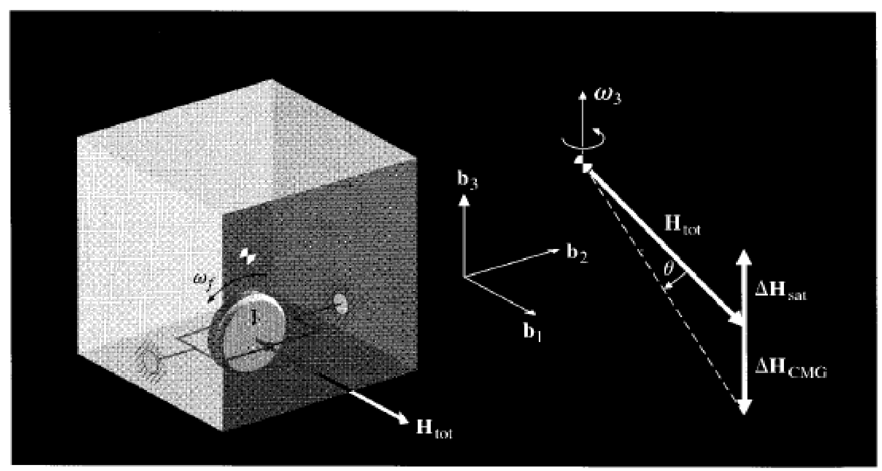
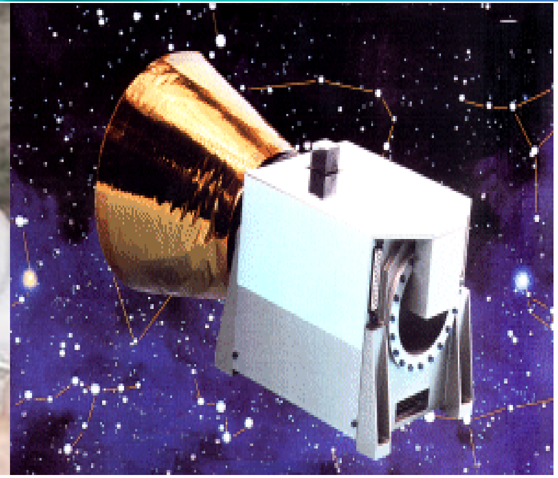


FIGURE 5.14
The control-moment gyroscope.

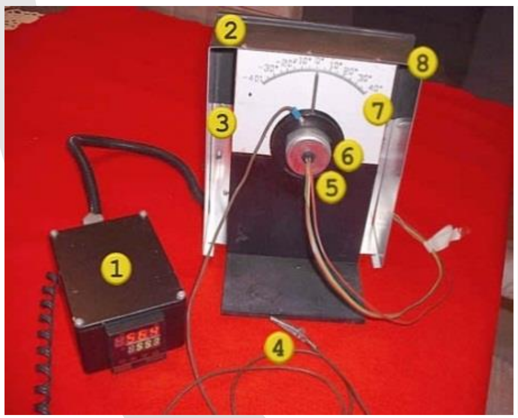
Control Momentum Gyro


Sun Sensor


Star Tracker


Thruster


Reaction Wheel



Earth Sensor


Magnetometer


Magnetic Torquer


Control Moment Gyro

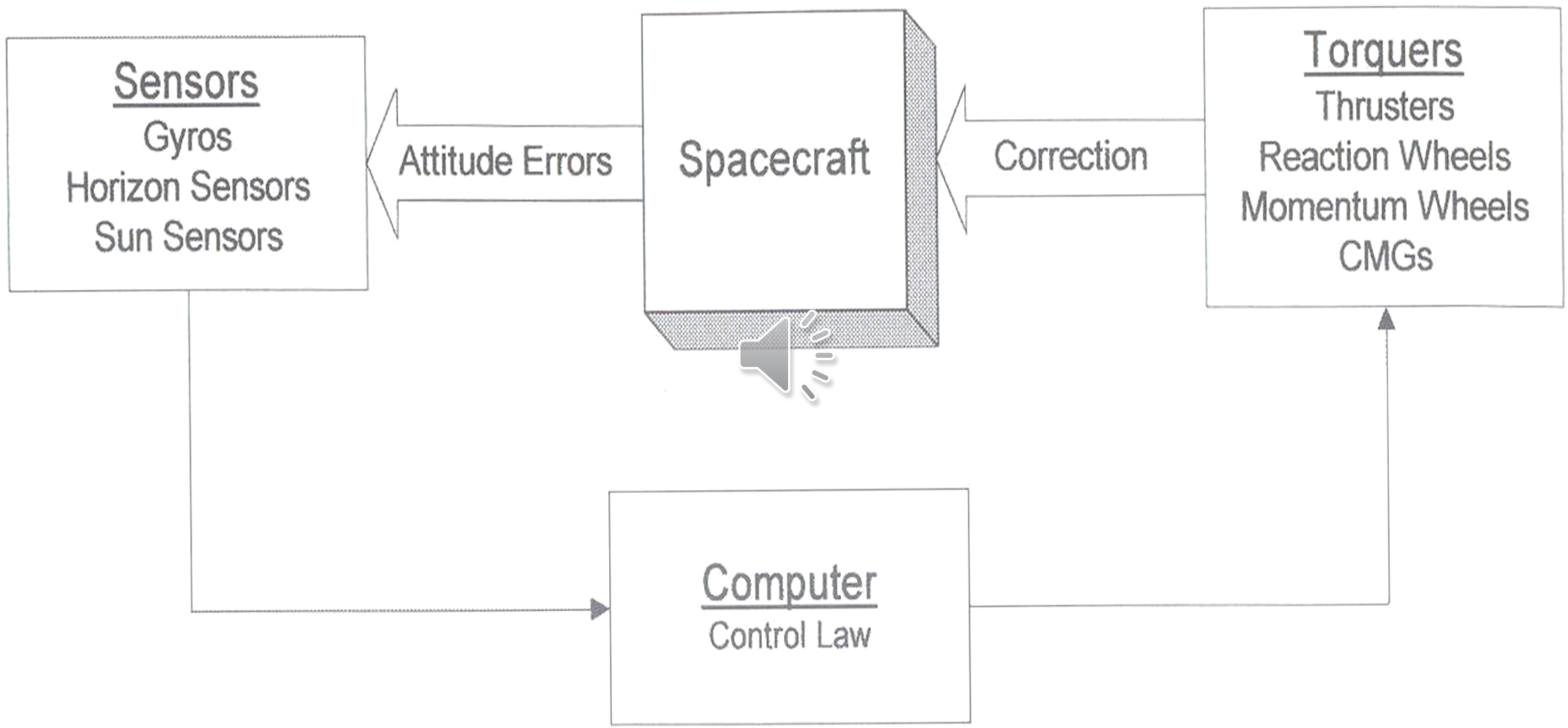


Fig. 5.4 Attitude control operation.

10.4 Designing the S/C Bus

TABLE 10-15. Types of Attitude Control.

Control Mode	Type of Control
Control During Thrusting:	
Spin Stabilization with Axial Thrust	Passive spin in a fixed direction with thrust applied parallel to the spin axis.
Spin Stabilization with Radial Thrust	Passive spin in a fixed direction with thrust applied perpendicular to the spin axis in short pulses.
3-Axis Control	Attitude is sensed with sensors whose output is used to control torquers. Torquers include thrusters operated off-on or swiveled to control thrust direction.
Control While Not Thrusting:	
Spin Stabilization with Precession Control	Spin direction is controlled by applying precession torque with an off-axis thruster.
Dual Spin	Spin-stabilized with a despun platform.
3-Axis Control	Control using attitude sensors and torquers.

10.4 Designing the S/C Bus

TABLE 10-16. Implication of Pointing Requirements on Attitude Control Approach.

Requirement	Implication
Control during kick-stage firing	Spin stabilization preferred in most cases
Coarse control (> 10 deg)	Spin stabilization or passive control using gravity gradient
Low-accuracy pointing (> 0.1 deg)	Either 3-axis control or dual spin
Low power requirement (< 1 kW)	Either oriented planar array or spinning cylindrical array
High-accuracy pointing (< 0.1 deg)	3-axis control
High power requirement (> 1 kW)	Oriented planar array
Multiple pointing requirements	3-axis control
Attitude slewing requirement	Dual spin with articulation mechanism or 3 axis with wheels

10.4 Designing the S/C Bus

- Accurate attitude control depends on the accuracy of attitude sensors.
- Attitude Control System can also produce attitude motions which combine with the attitude sensor's accuracy to affect the total control accuracy.

TABLE 10-17. Ranges of Sensor Accuracy. Microprocessor-based sensors are likely to improve accuracies in the future.

Sensor	Accuracy	Characteristics and Applicability
Magnetometers	1.0° (5,000 km alt) 5° (200 km alt)	Attitude measured relative to Earth's local magnetic field. Magnetic field uncertainties and variability dominate accuracy. Usable only below ~6,000 km.
Earth Sensors	0.05 deg (GEO) 0.1deg (low altitude)	Horizon uncertainties dominate accuracy. Highly accurate units use scanning.
Sun Sensors	0.01 deg	Typical field of view ±130 deg.
Star Sensors	2 arc sec	Typical field of view ±16 deg.
Gyroscopes	0.001 deg/hr	Normal use involves periodically resetting the reference position.
Directional Antennas	0.01deg to 0.5 deg	Typically 1% of the antenna beamwidth.

10.4 Designing the S/C Bus

- Estimate Torque Requirement
  - One Important sizing parameter for ADCS is the Torque Capability (Control Authority)
  - Must be large enough to counter balance disturbance torque & to control the attitude during maneuvers & transient events
  - Required torque to stabilize the S/C before it has exceeded a specified value

$$T = \frac{1}{2}\omega_t^2 (I_S / \theta_{\max}) \qquad (10-1)$$

T : Required Torque, ω_t : Tip-Off Rate

I_S : S/C Moment of Inertia, θ_{MAX} : Maximum Attitude Excursion

$$T = I_S \dot{\omega}_t = I_S \left(\frac{\theta_d}{t} \frac{1}{t} \right) = \frac{I_S}{t} \frac{\theta_d}{t} = \frac{I_S}{t} \frac{1}{t} \frac{\theta_{\max}}{2} = \frac{1}{2} \frac{\theta_{\max}}{t} \frac{\theta_{\max}}{t} (I_S / \theta_{\max}) = \frac{1}{2}\omega_t^2 (I_S / \theta_{\max})$$

  - Required torque to maneuver attitude by θ during t_{dur}

$$T = \frac{\Theta}{t_{dur}^{\ 2}} I$$

10.4 Designing the S/C Bus

- Estimate Angular Impulse for 3-Axis Control

I. Another major sizing parameter for ADCS is the angular impulse capability of its torque
   - Thruster Torque – Angular Impulse → Propellant Mass
   - Wheel/CMG – Angular Impulse → Wheel Moment of Inertia & Speed
   - Angular Impulse (Total Angular Momentum) required to start/stop an attitude maneuver

$$\boxed{L_{start} = I_S \omega_{man} \quad , \quad L_{stop} = I_S \omega_{man}}$$

II. The ADCS must counteract disturbance torques by applying control torque
   - Angular Impulse Capability > Disturbance Angular Impulse

10.4 Designing the S/C Bus

- Estimate Angular Impulse for 3-Axis Control (cont'd……)

TABLE 10-18. Disturbance Torques. These are vector equations where "×" denotes vector cross product and "•" denotes vector dot product. See Sec. 11.1 for simplified equations.

Disturbance	Equation	Definition of Terms
ΔV Thruster Misalignment	$\mathbf{s} \times \mathbf{T}$	$\mathbf{s}$ vector distance from center of mass to thrust application point $\mathbf{T}$ vector thrust
Aerodynamic Torque	$\frac{1}{2}\rho V^2 C_d A(\mathbf{u}_v \times \mathbf{s}_{cp})$	ρ atmospheric density C_d drag coefficient (typically 2.25) A area perpendicular to $\mathbf{u}_v$ V velocity $\mathbf{u}_v$ unit vector in velocity direction $\mathbf{s}_{cp}$ vector distance from center of mass to center of pressure
Gravity Gradient Torque	$\frac{3\mu}{R_0^3}\mathbf{u}_e \times (\mathbf{I} \cdot \mathbf{u}_e)$	μ Earth's gravitational coefficient $3.986 \times 10^{14}\ m^3/s^2$ R_0 Distance to Earth's center (m) $\mathbf{I}$ Spacecraft inertia tensor $\mathbf{u}_e$ Unit vector toward nadir
Solar Radiation Torque	$K_s(\mathbf{u}_s \cdot \mathbf{u}_n)A\begin{bmatrix}\mathbf{u}_s(\alpha + r_d) + \\ \mathbf{u}_n\left\{2r_s + \frac{2}{3}r_d\right\}\end{bmatrix} \times \mathbf{s}_c$	K_s solar pressure constant $4.644 \times 10^{-6}\ N/m^2$ $\mathbf{s}_c$ vector from spacecraft center of mass to area A $\mathbf{u}_n$ unit vector normal to A $\mathbf{u}_s$ unit vector toward the Sun α surface absorptivity coefficient r_s surface specular reflectance coefficient r_d surface diffuse reflectance coefficient (Note: $\alpha + r_s + r_d = 1$)

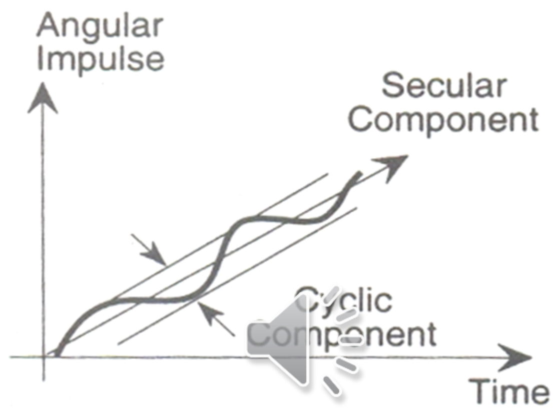
- Estimate Angular Impulse for 3-Axis Control (cont'd……)
  - III. Disturbance Torques may have cyclic & secular terms
    - Cyclic Terms integrate to zero for integer numbers of cycles
      - Wheel/CMG are appropriate
    - Secular terms are NOT periodic.
      - Thruster is appropriate
    - Use L of the 2nd step of Table 10-19 to size the thruster, CMG, etc

TABLE 10-19. Computing Control System Angular Impulse Requirements.

Step	Operation	Comments
1. Calculate disturbance torques	Use equations in Table 10-18	
2. Compute time integral of disturbance torque for each control axis	$L_x = \int T_{dx}\,dt$ $L_y = \int T_{dy}\,dt$ $L_z = \int T_{dz}\,dt$	L_x, L_y, L_z angular impulse requirements about x, y, and z control axes T_{dx}, T_{dy}, T_{dz} disturbance torques about x, y, and z control axes

10.4 Designing the S/C Bus

- Estimate Angular Impulse for 3-Axis Control (cont'd……)

3. Identify cyclic and secular components of angular impulse		Either identify cyclic components from the equations or plot each component of angular impulse to identify components
4. Size torquers		Wheels or CMGs are sized for the cyclic terms. Thrusters, if used with wheels, are sized for the secular terms.
5. If thruster control is used without wheels or CMGs, compute time integrals of the absolute value of disturbance torque	$L_x = \int \lvert T_{dx} \rvert dt$ $L_y = \int \lvert T_{dy} \rvert dt$ $L_z = \int \lvert T_{dz} \rvert dt$	

10.4 Designing the S/C Bus

- Estimate Angular Impulse for 3-Axis Control (cont'd……)

IV. Angular Impulse

- Thruster
  - Thruster Control Systems operate by pulsing a thruster when attitude error exceeds a set value known as the dead-zone limit, $2\theta_d$

$$P_{MIN} = Tst$$

P_{MIN} : Mimimum Impulse, T : Thrust Level

s : Lever Arm, t : Thruster Ejection Time $(0.02 - 0.1\,\text{sec})$

  - Average Impulse Rate = One Minimum Pulse every $2\theta_d/\omega$ second

$$IR = \frac{P_{\min}}{2\theta_d/\omega}\sec^{-1} = \frac{P_{\min}{}^{2}}{2\theta_d I_S}$$

  - Total Angular Impulse, $L = IR \cdot (Mission\ Duration)$

$$L = s\int T dt \cong sI_{sp} g m_p \quad \Leftrightarrow \quad m_p = L / sI_{sp} g$$

- Reaction Wheels $\quad L = I_\omega \Delta\omega_\omega$
- CMG $\quad L = I_\omega \Delta\omega_\omega (\Delta\theta)$

10.4 Designing the S/C Bus

- Estimate Angular Impulse for Spin Stabilization
  I. Angular Impulse for Spinup & Despin

$$L_{SPINUP} = I_{SS}\Omega_S, \quad \Omega_S : \text{Spin Speed } 0.1\text{-}1.0 \text{rad/sec}$$
$$L_{DESPIN} = I_{SS}\Omega_S$$

  II. Angular Impulse due to Thruster Misalignment

$$\Delta H = m_{S/C}\Delta v L_{cm}$$

  III. To change the direction of spin-axis orientation, we must precess the angular momentum vector

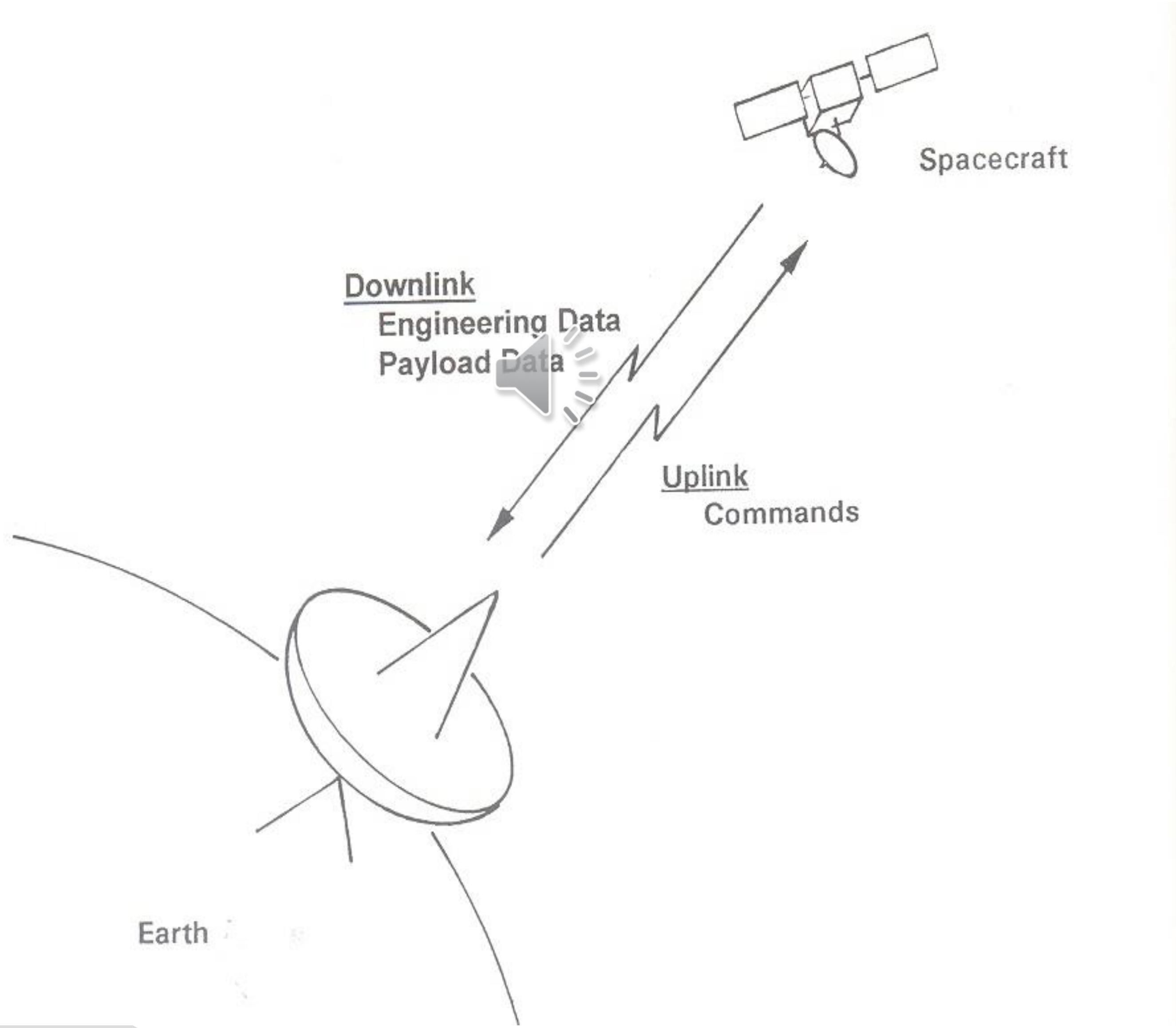
$$L_{precess} = I_{SS}\Omega_S\alpha, \qquad \alpha : \text{Angle of Rotation of Spin Vector}$$

TABLE 10-20. Weight and Power of Components in an Attitude Determination and Control Subsystem. Note $T \equiv$ Torque in N·m, and $H \equiv$ angular momentum in N·m·s.

Component	Weight (kg)	Power (W)
Earth Sensor	2 to 3.5	2 to 10
Sun Sensor	0.2 to 1	0 to 0.2
Magnetometer	0.2 to 1.5	0.2 to 1
Gyroscope	0.8 to 3.5	5 to 20
Star Sensor	5 to 50	2 to 20
Processors	5 to 25	5 to 25
Reaction Wheels	$2 + 0.4 \times H$ $H < 10$ $5 + 0.1 \times H$ $10 < H < 100$	10 to 20 at constant speed; 500 to 1,000 W/(N·m) when torquing
Control Moment Gyros	$35 + 0.05 \times H$ $100 < H < 2{,}500$	15 to 30 standby; 0.02 to 0.2 W/(N·m) when torquing
Actuators (single axis)	$4 + 0.03 \times T$	1 to 5 W/(N·m)

10.4 Designing the S/C Bus

10.4.3 Communication Subsystem



10.4 Designing the S/C Bus

- Consists of:
  - Transmitter
  - Receiver
  - Wide-beam Antenna
  - RF Diplexer (단향 이로 통신기)
- Receives and Demodulates Uplink Signals
- Modulates and Transmits Downlink Signals
- Command rate range is about 100bit/s ~ 100k bit/s
- Downlink signal consists of
  - Range Tones
  - Telemetry for S/C Status
  - Payload Data

- Data communication over a high bandwidth usually requires:
  - High-Gain
  - Directional Antenna
  - Low-bandwidth mode for wide-beam coverage

  - Bandwidth is the width of the range of frequencies included in the signal, and transmitted by the signal
- RF (radio frequency) links establish comm. between a transmitter and a receiver

10.4 Designing the S/C Bus

TABLE 10-21. System Considerations for Design of Communications Subsystems.

Consideration	Implication
Access	Ability to communicate with the spacecraft requires clear field of view to the receiving antenna and appropriate antenna gain
Frequency	Selection based on bands approved for spacecraft use by international agreement. Standard bands are S (2 GHz), X (8 GHz), and Ku (12 GHz)
Baseband Data Characteristics	Data bandwidth and allowable error rate determine RF power level for communications

TABLE 10-22. Steps in Designing a Communications Subsystem.

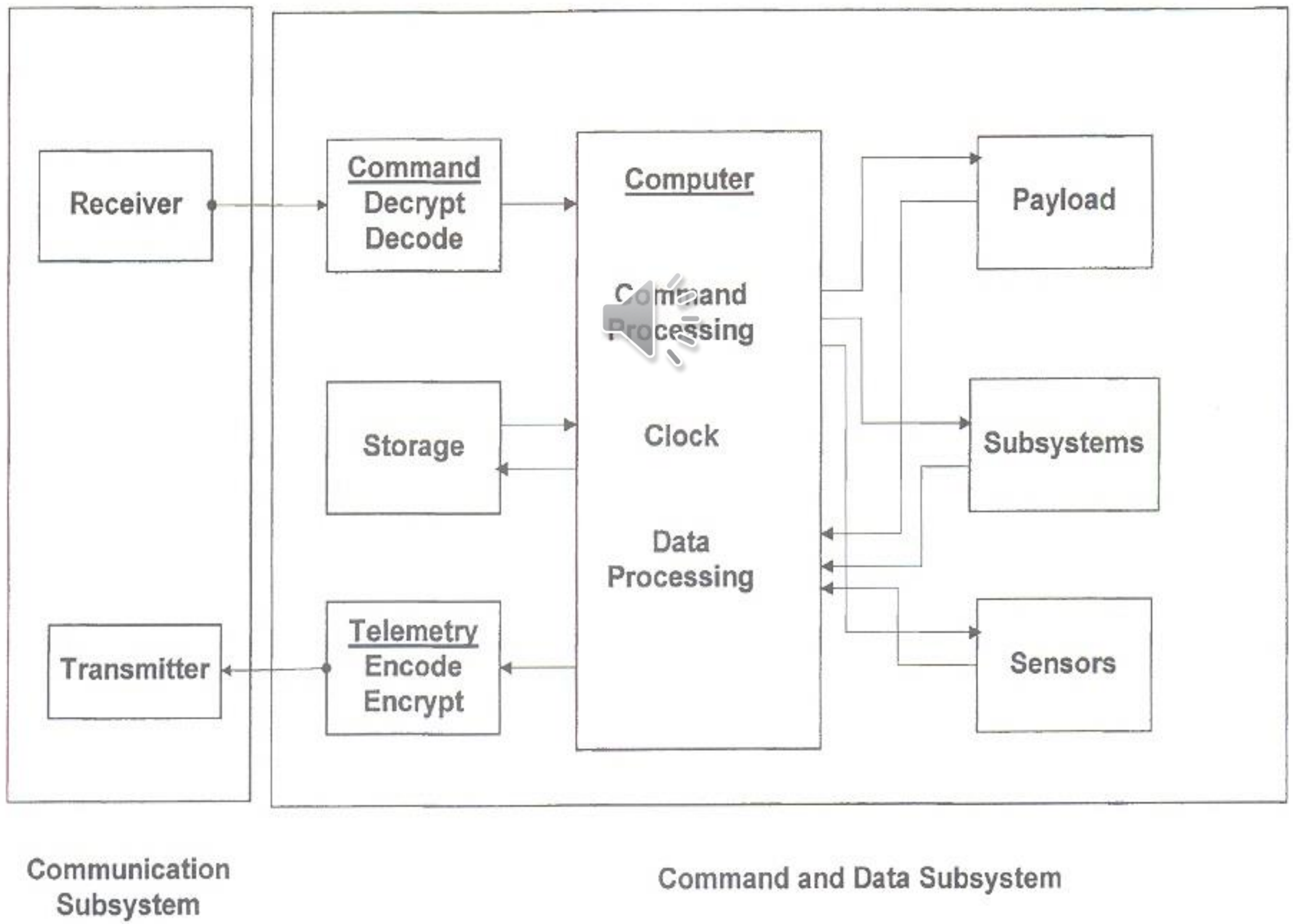
Step	What's Involved	Reference
1. Identify Data Rate		Payload commands and data—Chap. 9 Spacecraft bus commands and telemetry—Secs. 10.4.3, 11.3
2. Select Frequencies	Decide which of the allowed bands to use	Sec. 13.1
3. Prepare RF Power Budget	Analyze characteristics of RF links	Sec. 13.3
4. Select Equipment		Sec. 11.2

10.4 Designing the S/C Bus

TABLE 10-23. Characteristics of Communications Subsystems Using S-band.

Component	Weight (kg)	Power (W)	Comments
S-band Antenna	0.9	0	Hemispheric pattern 0 dB
Diplexer	1.2	0	
Receiver	1.8	4	Two units required for redundancy
Transmitter	2	4.4	

10.4.4 Command & Data Handling Subsystem



10.4 Designing the S/C Bus

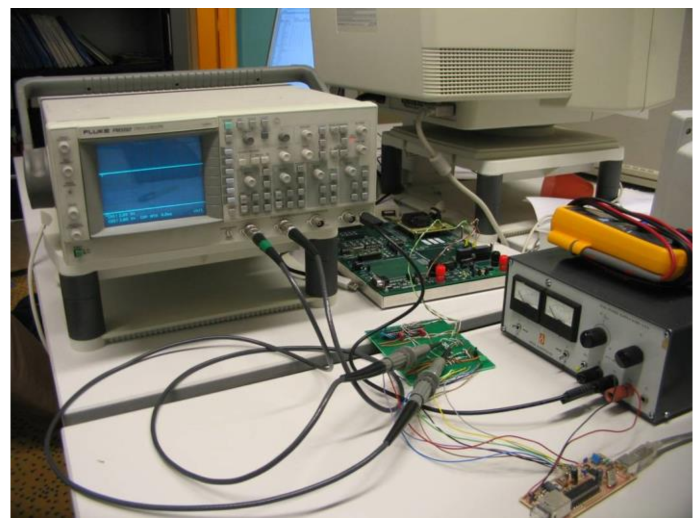
- Receives & distributes commands

- Collects, formats & delivers telemetry for standard S/C bus & payload operations
  - Telemetry signals provide S/C health and operational data needed to control the S/C.

- C&DH subsystem includes:
  - Encryptors (암호화 장치)
  - Decryptors (암호 판독기)
  - Sequencer (timer)
  - Computer
  - Data Storage Equipment

- C&DH interfaces with:
  - Comm. Subsystem
  - Payload
  - Other Subsystems

- Data Rate
  - House Keeping: < 1k bit/s
  - Payload: 10k~500M bits/s



10.4 Designing the S/C Bus

TABLE 10-24. Steps in Sizing the Command and Data Handling Subsystem.

Step	What's Involved	Reference
Prepare Command List	Prepare a complete list of commands for the payload and each spacecraft bus subsystem. Include commands for each redundancy option and each commandable operation.	Secs. 10.4, 11.3
Prepare Telemetry List	Analyze spacecraft operation to select telemetry measurement points that completely characterize it. Include signals to identify redundancy configuration and command receipt.	Secs. 10.4, 11.3
Analyze Timing	Analyze spacecraft operation to identify time-critical operations, and timeliness needed for telemetry data.	Sec. 10.3, Chap. 16
Select Data Rates	Choose data rates that support command and telemetry requirements and time-critical operations.	Sec. 13.3
Identify Processing Requirements	Examine need for encryption, decryption, sequencing, and processing of commands and telemetry.	Sec. 11.3
Identify Storage Requirement	Compare data rates of payload and spacecraft to the communications subsystem's ability.	Sec. 11.3
Select Equipment	Configure the subsystem and select components to meet requirements.	Sec. 11.3

10.4 Designing the S/C Bus

TABLE 10-25. Typical Characteristics of Basic Components for Command and Data Handling.

Component	Weight (kg)	Power (W)	Comments
Command Unit	5.0	5.4 standby 14 operating	Redundant unit, 9 user addresses capacity, 18,892 commands
Pulse Code Modulation Encoder	5.5	5.5	Redundant unit, 250 or 1,000 bits/s 64 word, 8 bit frame 5 subcommutated channels

10.4.5 Thermal Subsystem

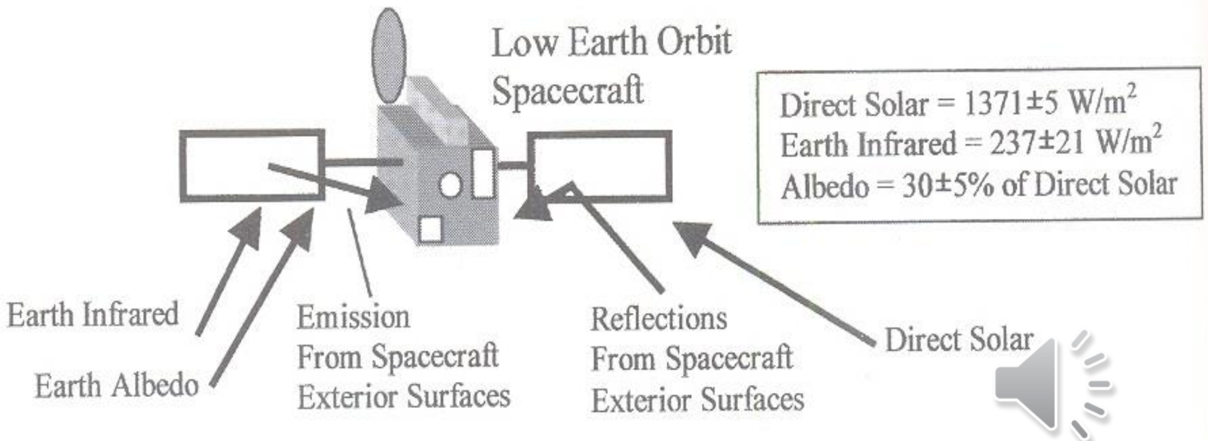


Fig. 7.1 Thermal radiation environment. (Values shown are from Ref. 1.)

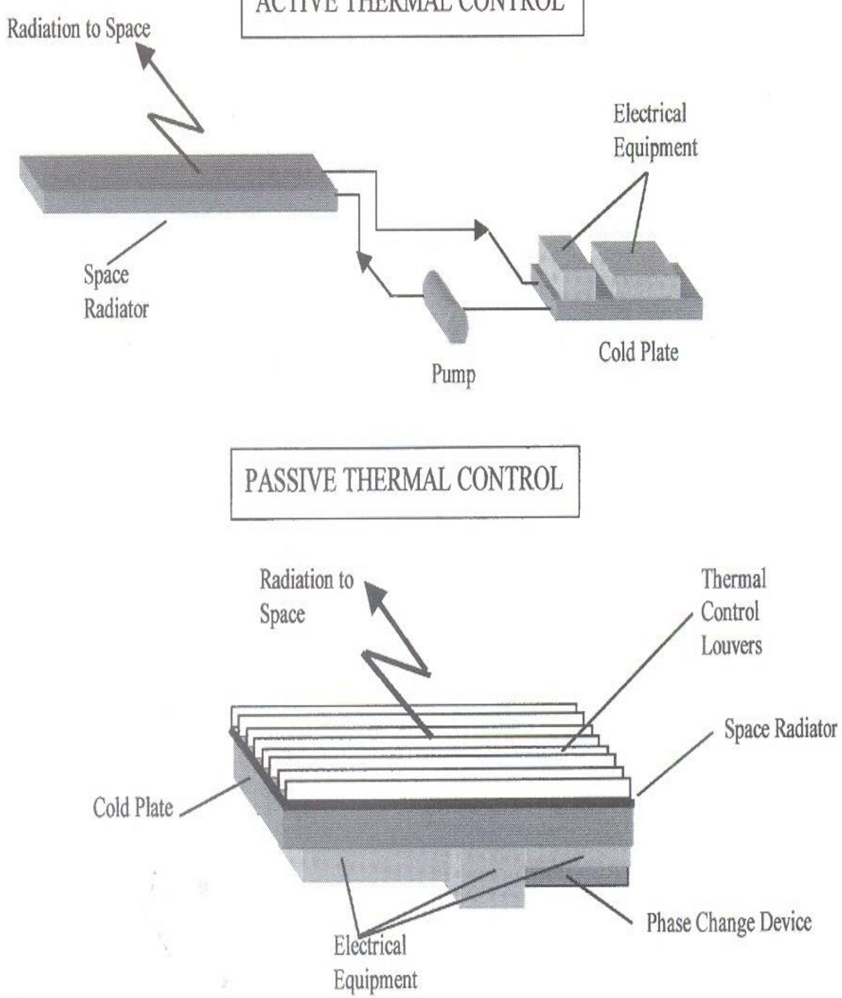


Fig. 7.2 Spacecraft thermal control subsystem arrangements.

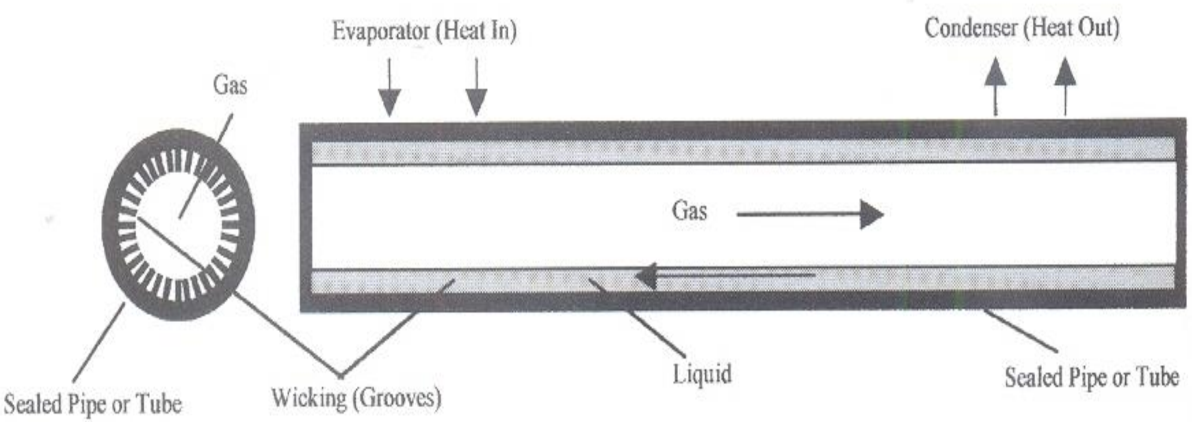


Fig. 7.3 Heat pipe.

10.4 Designing the S/C Bus

- Thermal Design of a S/C involves:
  - Identifying the sources of heat
    - Solar Radiation, Earth Reflection, Infrared Radiation, Electrical Energy from Electrical Components, etc.
  - Designing paths for transporting and rejecting heat
  - in order for components to stay within required temp.
    - Conventional electronics operate at 20℃
    - Battery cells are more sensitive to temperature (5~20℃)
- Process of Thermal Design
  - Identify heat sources and the location of radiating panels to dispose of excess heat
  - Compute the total amount of radiating area for the entire S/C

10.4 Designing the S/C Bus

- Heaters need most power for thermal control
  - Medium-sized S/C (100W) consumes 20W in the thermal subsystem

- Passive Control
  - Coating
    - Paints, Tape, Glass
    - Coating Weight < 4% of S/C Dry Weight)
  - Insulating
- Active Control
  - Heat Pipes to move heat from “hot spots”
  - Heater

10.4 Designing the S/C Bus

10.4.6 Power Subsystem

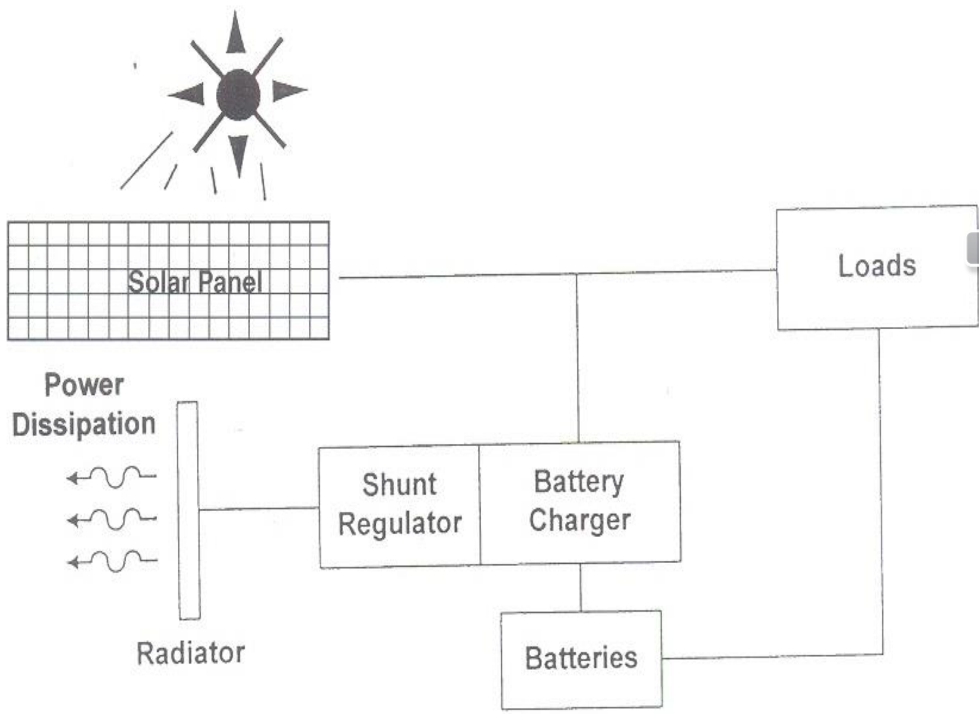


Fig. 6.2 Typical solar panel–battery power system (direct energy transfer architecture).

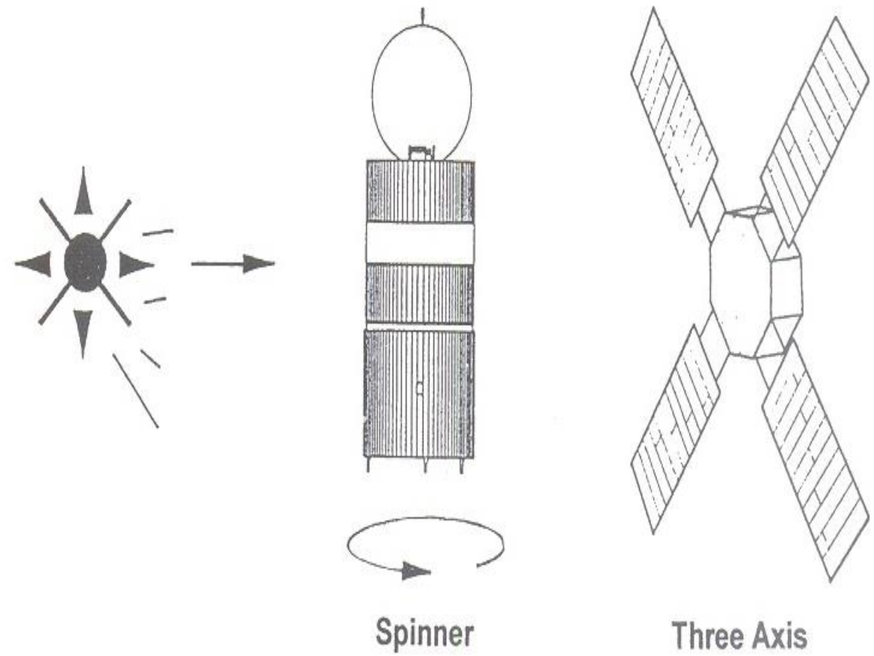


Fig. 6.4 Three-axis and spin-stabilized solar panels.

solar cells produce $1/\pi$ times, or about one-third, as much power as a sun-pointed panel of the same. High-power missions are difficult for a spinning spacecraft.

10.4 Designing the S/C Bus

- Power Subsystem
  - generates power
  - conditions and regulates power
  - stores power for periods of peak demand or eclipse operation
  - distributes power through S/C

  - Battery life limits the S/C lifetime

- Sizing the power subsystem is:
  - selecting a solar-array approach
  - sizing the array
  - sizing the batteries and the components that control charging
  - sizing the equipment for distributing and converting power

10.4 Designing the S/C Bus

- Solar Arrays
  - Planar Arrays
    - Flat panels pointed toward the Sun
    - Power is proportional to their projected area toward the incident sunlight
    - Often used for three-axis stabilized S/C
    - Required area of a planar solar array

      $$A = P/1367E$$

      A: area (m^2), P: required power (W), E: efficiencies (5~15%)
    - Mass of a planar array with specific performance $25W/kg$

      $$M = 0.04P$$

      M: mass (kg), P: required power (W)
  - Cylindrical Arrays
    - Appear on spin-stabilized S/C in which the spin axis is perpendicular or nearly perpendicular to the Sun line
    - Projected Area = $1/\pi$ of the total area
    - Should have π times as many cells as a planar array with the same power rating
  - Omnidirectional Arrays
    - can be used if S/C can receive sunlight from any aspect
    - Array must have equal projected area in all directions

10.4 Designing the S/C Bus

- Rechargeable NiCd or NiH_2 battery
  - Massive
  - Sensitive to temperature (5~20℃)

- Battery Capacity is determined from:
  - Energy it must produce
  - Depth-of-Discharge
    - select the battery's depth-of-discharge to meet cycle life requirements

TABLE 10-26. Allowed Battery Depth-of-Discharge vs. Cycle Life.

Cycle Life	Battery Type	Depth of Discharge
Less than 1,000 cycles	NiCd	80%
	NiH_2	100%
10,000 cycles	NiCd	30%
	NiH_2	50%

10.4 Designing the S/C Bus

- Ratio of Battery Weight to Capacity
  - NiCd: 20~40 $W \cdot hr/kg$
  - NiH_2: 35~50 $W \cdot hr/kg$

- Must match the solar array's electrical output to the battery's charging requirements
- Must provide switching equipment that allows the battery to supply power when needed
  → limiting battery's charge rate
  → limiting overcharge
  → providing for low-impedance discharge
  → providing for reconditioning

10.4 Designing the S/C Bus

- Power Control Unit's Weight for Battery
  - $\sim 0.02 kg/W$
- Power Dissipating Amounts
  - ~ 20% of the power converted
- Weight of Power Conversion Equipment
  - $\sim 0.025\ kg/W$
- Power dissipated in Wiring Losses and Switching Equipment
  - 2~5% of the operating power
- Wiring harness take up 1~4% of S/C dry weight

TABLE 10-27. Weight and Power Budget for Power Subsystem. P = required power in watts. Note that M_{dry} is used here as in Table 10-10.

Component	Weight (kg)	Power (W)	Comments
Solar Arrays	0.04 P	—	$\times \pi$ for cylindrical body-mounted $\times$ 4 for omnidirectional body mounted
Batteries	C/35 (NiCd) C/45 (NiH_2)	—	C = capacity in W·hrs
Power Control Unit	0.02 P	—	P = controlled power
Regulator/Converters	0.025 P	0.2 P	P = converted power
Wiring	0.01–0.04 M_{dry}	0.02–0.05 P	M_{dry} = spacecraft dry weight

10.4.7 Structures and Mechanisms

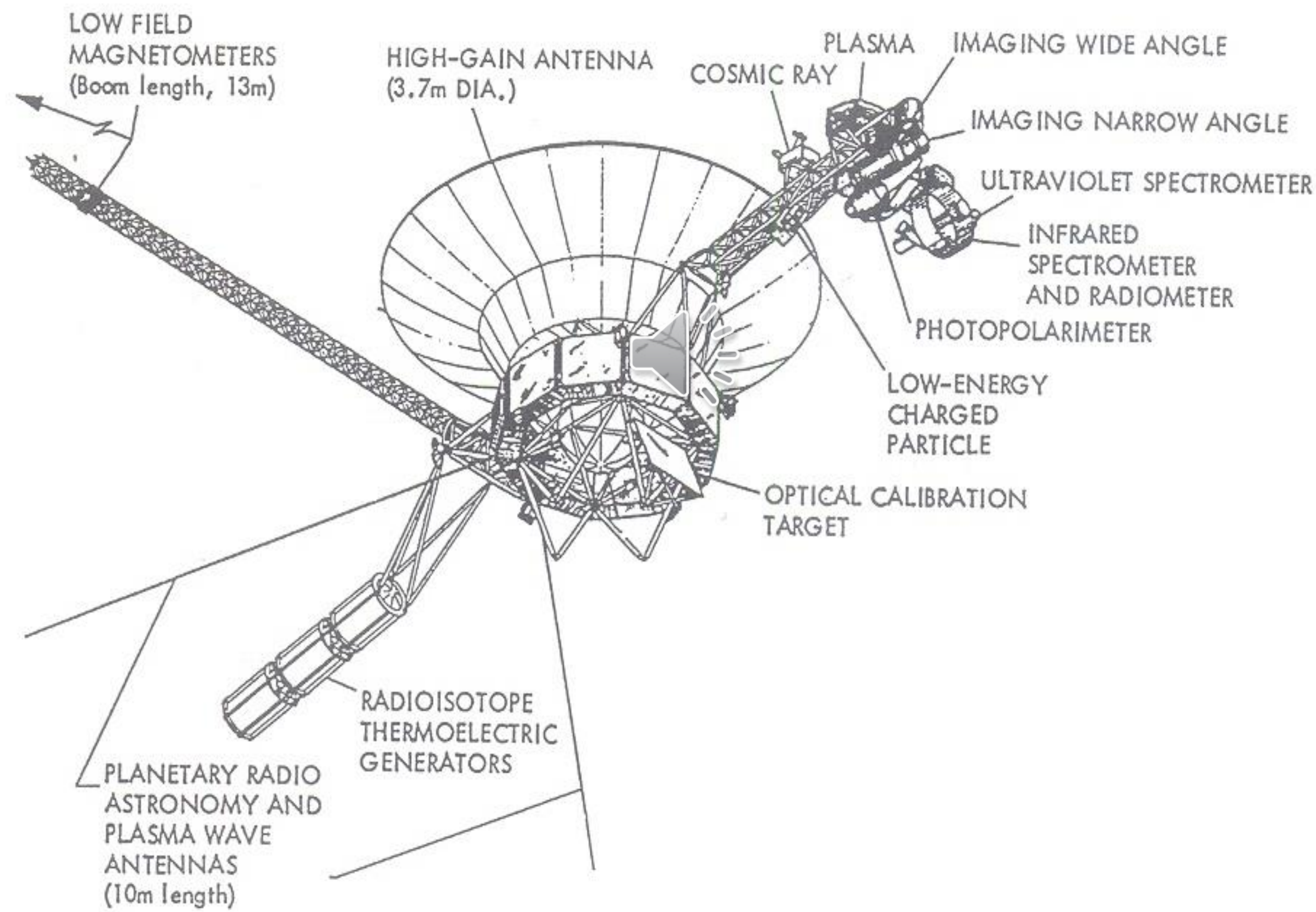


Fig. 10.3 Voyager spacecraft. (Reproduced courtesy of NASA/JPL/Caltech; Ref. 10, p. 17.)

10.4 Designing the S/C Bus

- Primary Structure
  - Carries and protects S/C and payload equipment through the launch environment
  - Size is based on launch loads (stiffness, strength)
    - 6g's axial acceleration
    - 3g's lateral acceleration

- Secondary Structure
  - Deploys S/C after orbit injection
  - Size is based on on-orbit factors

- Structural mass of 10~20% of S/C dry mass is a reasonable starting point